

Business Valuation — Study Notes

ICAI Advanced Financial Management, Chapter 13 (68 pages condensed)

0. Chapter map (what to remember at a glance)

#	Topic	One-line takeaway
1-2	Framework & terms	Valuation = science + art; used for M&A, listing, strategy, ball-park price
3	3 core approaches	Asset, Income/Earnings, Cash flow (DCF)
4	Cost of equity	CAPM, APT, un-levering/re-levering beta for unlisted cos.
5	Relative valuation	Multiples of comparable firms (EV/EBITDA, P/E, PEG...)
6	Other approaches	LBO, Chop-shop, EVA, MVA, SVA
7	Fair value	Give a <i>range</i> (seller's minimum – buyer's maximum)
8	Going vs non-going concern	Going concern value > liquidation value
9	Distressed cos.	Modified DCF, DCF + distress value, APV, relative
10	Start-ups	Berkus, Scorecard, VC, First Chicago, Cost-to-duplicate, Comparable txns
11	Digital platforms	Income / Market (revenue-driver based) / Cost approach
12	Professional firms	Normalise earnings, KPI/benchmark comparison
13	ESG	Adjust discount rate <i>or</i> (better, more explicit) cash flows

1. Conceptual framework

Why value a business?

(a) Info for internal stakeholders (b) compare with similar enterprises / mgmt efficiency (c) future public listing (d) strategic planning – value drivers, deployment of surplus cash (e) ball-park price for acquisition.

- Valuation is both **science and art**.
- Listed target → market cap is a guide **but** depends on market efficiency; sometimes only a small float is traded. Unlisted target → negotiated price.
- (Historical trivia: the East India Company was the first corporation valued / IPO'd.)

2. Important terms

Term	Meaning / formula	Notes
PV	$CF / (1+r)^n$	₹1,000 in 1 yr @10% ≈ ₹909
IRR	Rate at which $PV(\text{inflows}) = PV(\text{outflows})$, i.e. NPV = 0	Higher IRR → project more likely selected
ROI	$(\text{Gain} \div \text{Investment})$	Eg. $(1400-1000)/1000 = 40\%$. With additional investment use average investment : returns $800 \div \text{avg}(1000,1200)=1100 \rightarrow 72\%$. ROI is a <i>historical</i> ratio

Term	Meaning / formula	Notes
Gordon (perpetual growth) model	$P_0 = D_1 / (k - g)$	Little used in practice (dividends rarely grow perpetually, many parameters) – academically neat
TV (Terminal / Horizon Value)	Value at the exit/horizon; usually Gordon : $CF_n \times (1+g) / (WACC - g)$	In DCF, TV is often the <i>largest</i> part of value → validate g carefully

3. Three approaches to valuation

3.1 Asset-based approach

Value of shares = net assets acquired. Not income-based. **Least important for IT/knowledge companies** (assets are IP and people).

Method	Definition	Points
Net Asset Value (Book value)	Fixed assets + Net current assets – Long-term debt	Simplest; uses historical cost; ignores brand/IPR → gives lower limit
Net Realisable Value (Liquidation / Adjusted book value)	Realisable value of all assets – liquidation expenses – liabilities	Used when acquirer will sell one part & integrate rest; involves total break-up; buyer likely to offer lowest price
Replaceable Value	Cost of acquiring equivalent assets/liabilities in open market	Slightly higher than NRV; often seen as max price an acquirer would pay; ignores staff loyalty

Example (NRV): Liabilities 20,000. Book: PPE 50,000 + Licences 10,000 + Debtors 50,000 + Cash 10,000 → net 1,00,000. Realisable: PPE 40,000, Licences 30,000, Debtors 45,000, Cash 10,000 → net **1,05,000** → ₹21/share (5,000 shares).

Conclusion: reflects net worth under going concern principle **but ignores ability to generate future revenue / market dynamics.**

3.2 Income-based approach

Overcomes asset-approach drawbacks by using earning potential; suitable when acquirer will **continue** the business.

1. PE ratio / Earning yield multiplier

- **Price/share = EPS × PE ratio**
- For unlisted cos.: (i) choose PE of comparable listed co. (ii) **adjust downward** for non-listing risk (iii) determine future maintainable EPS (iv) multiply.
- Gives *minimum acceptable price* to target shareholders.

2. Capitalisation of earnings

- **Capitalised value = Expected annual maintainable profit ÷ Capitalisation rate**
- Cap rate ≈ **EPS / Share price = 1 / PE ratio** (or CAPM-based).
- Maintainable profit = weighted average of past profits adjusted for synergies/economies of scale.
- *Pro:* forward looking. *Con:* forecasting profit; treatment of extraordinary items.

3.3 Cash-flow (DCF) approach - 5 steps

(a) Arrive at FCF → (b) forecast future FCFs → (c) discount rate (WACC) → (d) Terminal Value → (e) PV of FCFs + PV of TV, interpret.

Worked example (₹'000)

- FCF: EAT 600 – one-time income 200 + one-time expense 100 + depreciation 100 = **600**
- Grow 5% p.a.: 600 / 630 / 661.5; less Δ WC and FA investment → **Adjusted FCF 500 / 550 / 651.5**
- WACC 8%; PVF 0.926 / 0.857 / 0.794 → PV **463 / 471.35 / 517.29**
- TV = 517.29 × 1.03 / (0.08 – 0.03) = **10,656.17** (the book uses the Year-3 PV figure here and adds it undiscounted)
- Total value = PV of flows (1,451.64) + TV 10,656.17 = **12,107.81 ≈ 12,108** (₹'000)

⚠ Exam tip: DCF is intrinsic and forward looking, but results hinge on **WACC and TV** assumptions
– TV is often the biggest share of value, so validate g.

4. Measuring cost of equity

4.1 CAPM

$$R = r_f + \beta (r_m - r_f)$$

r_f = risk-free rate, β = beta, r_m = market return. Intuitive (risk-free part + market-relative risk) but **over-simplifies risk**.

4.2 Arbitrage Pricing Theory (APT) — Stephen Ross, 1970s

$$R = r_f + \beta_1(RP_1) + \beta_2(RP_2) + \dots + \beta_n(RP_n)$$

Multi-factor: each macro-economic factor (interest rates, sector growth etc.) has its own beta/risk premium. APT doesn't say *which* factors – analysts must choose according to the economy.

4.3 Beta for unlisted companies / new businesses

- Listed cos. usually have debt; private firms may have little → need **unlevered (asset) beta**.
- A listed company entering a new line shouldn't use its own WACC; use **asset beta of proxy (pure-play) firms**, re-gear to own capital structure.
- **Asset beta = business risk only** (no financial risk). Equity beta $\geq$ asset beta (equal if debt-free).

Formulas

- $\beta_a = \beta_e \times [E / (E + D(1-t))] + \beta_d \times [D(1-t) / (E + D(1-t))]$
- If **debt beta = 0** (usual assumption): $\beta_a = \beta_e \times E / (E + D(1-t))$
- **Re-gearing:** $\beta_e = \beta_a \times [E + D(1-t)] / E$

5 steps: (1) identify pure-play/proxy firms & their equity betas → (2) de-gear to asset betas → (3) average (or pick the most appropriate) → (4) re-gear to appraising company's capital structure → (5)

plug β_e into CAPM → required return.

Illustration 1 - X Pvt Ltd (retail, angel investors)

Data: EBITDA 90 (includes ₹10 extraordinary gain; ₹20 pending write-off), unlevered β 1.8, D:E = 40:60, r_f 5%, r_m 11%, EV multiple 5x, pre-tax Kd 12%, tax 30%, FCF Y1-3 = 100/120/150.

- Levered $\beta = 1.8 [1 + (1-0.3)(40/60)] = \mathbf{2.64}$
- Adjusted EBITDA = $90 - 10 - 20 = \mathbf{60} \rightarrow EV = 5 \times 60 = \mathbf{300}$
- $K_e = 5\% + 2.64 \times 6\% = \mathbf{20.84\%}$
- $WACC = 20.84 \times 0.6 + 12(0.7) \times 0.4 = \mathbf{15.864\%}$
- Discount factors 0.863 / 0.745 / 0.643 → PVs 86.30 + 89.40 + 96.45 = **272.15** (value of firm)

5. Relative valuation (valuation by multiples)

- Intrinsic methods (§3) give *intrinsic* value; relative valuation gives value **relative to peers**, increasingly used to **validate** intrinsic value.
- Idea: derive a "multiple" from comparable firms' financial ratios & apply to target.

4 steps

1. **Pick the driver / multiple**
 - *Enterprise-value multiples*: **EV/EBITDA** (popular), **EV/Invested capital** (capital-intensive), **EV/Sales** (cash-rich, big order book, organic growth)
 - *Equity multiples*: **P/E**, **PEG = P/E ÷ growth rate** (best for high-growth / sunrise industries)
2. **Choose the right ratio** (factor-based thinking: eg. an exporter is affected by FX; EV/Invested capital is a misfit for an asset-light firm)
3. **Choose comparable firms** – hardest step; no two firms identical (eg. FTE pricing vs. unit-transfer pricing changes risk). If niche, look at other industries or use **regression**. "*Take benchmarks with a pinch of salt.*"
4. **Iterate / extrapolate** to smooth deviations.

Assumptions: market is efficient; multiples-fundamentals relationship is linear; comparables are similar in structure, risk and growth.

Enterprise Value (EV) = Market cap + Debt – Cash (two routes: entity-value base adjusted for debt, or balance-sheet approach).

Illustration 2 - A Ltd.

Gross profit 10,00,000; indirect exp 4,00,000 → EBITDA 6,00,000; 1,00,000 shares; debt 3,00,000; surplus funds 5,00,000; r_f 4.5%, r_m 12%, β 0.9.

- Cap rate = $4.5 + 0.9 \times 7.5 = \mathbf{11.25\%}$
- Earning value = $600 / 0.1125 = ₹5,333.33$ thousand → **₹53.33/share**
- Equity value @ 5x EBITDA: $3,000 - 300 \text{ debt} + 500 \text{ surplus} = \mathbf{₹3,200 \text{ thousand}}$

Illustration 3 - HK Ltd. (EV via balance sheet)

70,000 shares $\times$ ₹12 = ₹8,40,000 market cap + debt 2,00,000 – cash 5,00,000 = **EV ₹5,40,000**.

6. Other approaches to value measurement

6.1 Contemporary approaches

- Industry-specific metrics: internet cos. $\rightarrow$ *price per page visited*, *price per subscriber* (still check cash-to-sales / DCF).
- **Goodwill-based:** asset valuation first, then goodwill separately via multiple of annual sales / footfall (e.g., retail hotspot).
- **P/E ratio (PER):** EPS vs. market price (EPS 40, price 50 $\rightarrow$ 1.25 per the book's text). Relative figure – compare within same sector.
- **LBO (Leveraged Buy-Out):** PE firms buy using **~70:30 debt:equity**, hold 3–5 yrs, then spin off / sell. Risk: future can't be predicted – a cyclical slowdown leaves heavy debt/interest.
- *Example:* Software co. EBITDA ₹100 L, VC buys majority (consideration ₹250 L), 20% growth/yr; exit at **7 $\times$ EBDAT**. EBDAT Y3 = 112 $\rightarrow$ capitalised 784 – debt 100 = **Equity ₹684 L**.

6.2 Chop-shop (break-up value) method

Finds multi-industry firms that are **undervalued** and worth more separated (assets bought below replacement value).

1. Identify segments & average capitalisation ratios of pure-play firms.
2. Compute "theoretical" value of each segment on each ratio (cap/sales, cap/assets, cap/operating income).
3. **Average** the theoretical values.

Illustration 4 - Cornett GmbH: values on sales = €4,005,000; on assets = €2,880,000; on income = €5,650,000 $\rightarrow$ **average €4,178,000** (vs. market cap €4m $\rightarrow$ fairly valued/slightly undervalued).

6.3 Economic Value Added (EVA)

- Company creates value only if returns **exceed cost of capital**. Separates operations from financing. Judges *management performance*, not just numbers.
- **EVA = NOPAT – (Invested Capital $\times$ WACC) = NOPAT – Capital charge**
- **NOPAT** = EBIT – taxes (depreciation not added back; adjust other **non-cash** items e.g. bad-debt provision).
- **Invested capital** = total assets – non-interest-bearing current liabilities ($\approx$ equity + long-term debt, usually opening), adjusted for non-cash items.
- Negative EVA $\Rightarrow$ not creating value.

Illustration 5 - A Ltd. (₹ lakh)

EBIT 410; tax 123 (30% $\times$ EBIT); add back bad-debt provision 20 $\rightarrow$ **NOPAT 307**

Invested capital = 1300 – 400 + 20 = **920**

$$WACC = (800/900)(8.45\%) + (100/900)(12\%)(0.7) = \mathbf{8.44\%}$$

$$\text{Capital charge} = 920 \times 8.44\% = 77.65 \rightarrow \mathbf{EVA = 307 - 77.65 = 229.35}$$

6.4 Market Value Added (MVA)

MVA = Market value of the firm – Invested capital (e.g. 1000 – 920 = **80**).

- Market's perception of value created; EVA is the *derived* value added for the discerning investor.
- High MVA companies = market darlings (expensive); short-term EVA/MVA can be negatively correlated, gap closes eventually.
- **Objective of EVA:** show management efficiency in earning above the hurdle rate.

6.5 Shareholder Value Analysis (SVA)

Fills NOPAT's gap (historical, ignores future cash flows & investment opportunities). Key **value drivers**: sales/earnings potential, investment opportunities, cost of incremental capital.

Steps: (a) FCFs from value drivers → (b) discount at WACC → (c) add TV → (d) add market value of non-core assets & marketable investments → (e) subtract debt = **Value of equity**.

Case study (\$ m): PV of FCFs (57.54 + 43.46 + 44.86) + PV of TV 336.95 = **482.81**; + investment property 35 – debt 19 = **Equity value 498.81**.

(Y4-onwards FCF 63.20 × PVF 0.64 = 40.45; × TV multiplier 1/0.12 = 8.33 → 336.95.)

Quick comparison

	EVA	MVA	SVA
Basis	Accounting (NOPAT – capital charge)	Market value – invested capital	Forward-looking DCF with value drivers
Looks at	Management efficiency (past/current)	Market's perception	Future value creation
Weakness	Historical, ignores value drivers	Not a basis for share valuation	Forecast dependent

7. Arriving at fair value

- Acquirer wants to pay the fair price – "no less, no more". Fair value = arm's-length transaction (CA view); PV of cash (analyst view); arbitrage opportunity (speculator view).
- Booms distort value (dot-com).
- No single "correct" method → determine a **range: minimum acceptable to seller – maximum payable by buyer**; final price by negotiation.
- **Exam approach for evaluation/synthesis questions:** (i) unless specified, value by **as many methods as data permit** (ii) **comment** on each method's result (iii) supplement the conclusion with any additional information.

8. Going concern vs non-going concern

- Going-concern assumption: enterprise continues for foreseeable future (no intention/need to liquidate).
- **Non-going concern = liquidation value** (net value after selling assets and paying liabilities).
- **Going concern value (Total value) > liquidation value** because it includes future profitability, intangibles, goodwill; liquidation also hurts reputation and lays off employees.
- Use non-going-concern valuation **only** when investors believe the firm has no future value as a going concern.

9. Valuation of distressed companies

Distress = unable (or struggling) to meet financial obligations – high fixed costs, illiquid assets, cyclical revenue; too much debt; inability to meet operating expenses. A distressed firm is **not necessarily worthless**.

Why conventional methods fail

- DCF's TV assumes infinite life & growth; flows may be negative.
- Negative/declining revenues → cash flows hard to estimate, high bankruptcy risk; DCF treats firm as going concern.
- Discount rates reflect healthy firms → need adjustment for probability of failure.

Methods

1. **Modified DCF:** estimate probability distribution of cash flows incl. default; adjusted discount rates (updated D/E and unlevered β for K_e ; updated default risk for K_d).
*If distribution can't be estimated: **Expected CF_t = CF_t × (1 – Probability of distress_t)***
2. **DCF + Distress value:**
Equity value = DCF equity value × (1 – P(distress)) + Distress sale value of equity × P(distress)
(distress sale value as % of book value or % of DCF value)
3. **Adjusted Present Value (APV):** separates investment & financing decision.
Firm value = Unlevered firm value + (PV of tax benefits of debt – Expected bankruptcy cost)
Expected bankruptcy cost = (Unlevered firm value – Distress sale value) × Probability of distress
4. **Relative valuation:** revenue/EBITDA multiples more usable than P/E, P/B; adjust the multiple downward (eg. industry 2× revenue → 1.25× for the distressed telecom).

10. Valuation of start-ups

- Standard approaches (earnings/CF, asset, market) presuppose an established, profitable business with history. Start-ups have **no track record**; value = future growth potential, founder skill ("night-vision goggles").

Why traditional methods fail

| Approach | Reason |

|-----|-----|

| Income | Most start-ups have no positive cash flow in the foreseeable future |

| Asset | Few tangible assets (value is IP/intangibles, hard to agree); going-concern so value $\neq$ realisable asset value |

| Market | Disruptors $\rightarrow$ no established comparables; (but some market-approach elements used in later funding rounds) |

Value drivers: Product (readiness/prototype), Management (credentials & balance of team), Traction (quantifiable evidence of demand), Revenue streams, Industry attractiveness (logistics, distribution, lockdown risk...), Demand-supply of investors, Competitiveness (first-mover advantage vs. proven model).

Methods

| Method | Gist |

|-----|-----|

| **Berkus** (Dave Berkus) | Rate 5 success factors: **basic value (idea), technology, execution, strategic relationships, production & sales**. Caps pre-revenue value at **\$2m**, post-revenue at **\$2.5m** (as per text) |

| **Cost-to-duplicate** | Sum of all costs to build the start-up/product incl. physical assets. Criticised: ignores future revenue/assets |

| **Comparable transactions** | Value similar start-ups sold at. XYZ acquired ₹560 cr with 24 cr users $\approx$ ₹23/user $\rightarrow$ ABC with 1.75 cr users $\approx$ **₹40 cr**; adjust multiplier for tech, IP, penetration, location |

| **Scorecard** | Take avg pre-money valuation of comparables; weight factors - **team 0-30%, opportunity size 0-25%, product 0-15%, competition 0-10%, marketing/sales/partnerships 0-10%, need for additional investment 0-5%, others 0-5%**; assign comparison % (100% = par); sum of weighted factors $\times$ avg valuation = pre-revenue value |

| **First Chicago** | DCF + market approach across **3 scenarios: worst, normal, best**, each probability-weighted |

| **Venture Capital method** | Investor wants multiple (10 $\times$, 20 $\times$, 30 $\times$) or target IRR on exit; discount exit/future value to get **post-money value** at required return |

11. Valuation of digital platforms

Digital platform = software-based online infrastructure that facilitates many-to-many interactions/transactions.

Category	Eg.	Revenue drivers (market approach)
Marketplace (supply-demand)	Booking.com, Uber, Amazon	bookings, registered users, transaction volume
Search engine	Google, Bing, Baidu	active users, relevance of results, time per search
Repository (library)	Spotify, YouTube, GitHub	readers/contributors, authenticity, duration of use, content quality
Digital communication	WhatsApp, Teams, Telegram, Slack	users, sponsored links, ad revenue
Digital community	Facebook, LinkedIn	users, subscription fees

Category	Eg.	Revenue drivers (market approach)
Payments	Paytm, GPay	active subscribers, merchants, speed/security/ease of use

Income approach

- *Top-down*: start with **TAM** (total addressable market) → **SAM** (serviceable addressable) → **SOM** (serviceable obtainable) → business plan.
- *Bottom-up*: build from the platform's own resources/earnings (better for nascent platforms).
- Platforms burn cash early (penetration pricing); cash need falls as margins stabilise.
- **Discount rate**: FCFF → **WACC**; FCFE → **Ke** (CAPM). Beta hard (few listed comparables) → use sector/revenue-driver/international comparables, **add a Company-Specific Risk Premium (CSRP / alpha)** – higher for nascent cos.

Market approach: P/E, EV/EBITDA, P/B, **Price/Revenue**. Difficult because: comparables scarce, profit/EBITDA may be negative, capital-light → low book value. So emphasise **revenue drivers** (users, views per user, revenue per user).

Cost approach: cost to rebuild (developer hours on code). Ignores revenue-generating power – usually not most appropriate.

Fundamentals unchanged: understand the business, revenue model, management quality, risk-reward.

12. Valuation of professional / consultancy firms

- Firms providing customised, knowledge-based services (CAs, advocates, consultants).
- Compare historical data with **industry KPIs and benchmarks** and competitors; sources: audited annual statements, ITRs.
- Income approach: historical data + projected growth (TV) – factor the **risk that projections don't materialise**.
- **Normalise net income and cash flows** (add back non-cash and non-recurring/owner-specific items) so firms are compared on an equal footing; then apply chosen valuation method.
- Pick KPIs aligned with the **acquirer's goals**.

13. Impact of ESG on valuation

E	S	G
Climate change, water, waste, emissions, biodiversity	Employee development, diversity & inclusion, community development, health & safety, customer	Board independence & diversity, anti-corruption & bribery, tax transparency, ethical conduct

- Momentum: ESG funds, green bonds (> \$1 trillion in 2020), sustainability taxonomies (EU), single global ESG standard (IFRS/ISSB), **SEBI Feb 2023** proposed ESG disclosure framework.
- Benefit of high ESG: preferential/lower cost of debt, access to green/social/sustainability-linked bonds. ESG has moved from "good to have" to "must have".

- **How to incorporate in valuation (DCF):** adjust either **discount rate** (add risk premium – practical but less explicit) **or expected cash flows** (more explicit – preferred):
- **E:** 2-degree scenario analysis; carbon points/prices
- **S:** cost of social measures – labour conditions, CSR, welfare measures
- **G:** penalties, fines, taxes from poor governance

14. Case studies

1. **Vodafone-Idea merger:** valuation by (a) market value (NSE price), (b) comparable companies' multiples, (c) NAV (as of 31-12-2016). **DCF was not used** (management didn't provide projections). Share-exchange basis. Market reaction: Idea fell ~9.6% (as much as 14.57%) – the deal was thought to undervalue Idea. **Lesson:** technically correct valuation can still disappoint – "perceived value" is not quantifiable.
2. **Facebook-WhatsApp (2015, \$21.8 bn):** free service (\$1/yr fee), 450 m users (70% active), growth 1 m users/day → ~\$55 gross per user (incl. \$4 bn retention payout). Value rests on users/future monetisation (cross-platform international messaging).

15. Formula sheet (revision)

Item	Formula
Price per share (PE)	EPS × PE
Capitalised value	Maintainable profit ÷ cap rate (cap rate = 1/PE)
CAPM	$K_e = r_f + \beta(r_m - r_f)$
APT	$R = r_f + \sum \beta_i RP_i$
Asset beta (Kd beta = 0)	$\beta_e \times E / [E + D(1-t)]$
Re-gear beta	$\beta_a \times [E + D(1-t)] / E$
WACC	$K_e \cdot E / (E+D) + K_d(1-t) \cdot D / (E+D)$
FCF	EAT ± one-offs + Dep – ΔWC – Capex (as per question)
Terminal value	$FCF_n(1+g)/(WACC-g)$ ($FCFF_1/(K_c-g)$ for FCF in next year)
EV	Market cap + Debt – Cash
PEG	P/E ÷ growth rate
NOPAT	EBIT(1–t)
EVA	NOPAT – WACC × Invested capital
MVA	Market value – Invested capital
EVA dividend per share	EVA ÷ no. of shares (max dividend before value starts to fall)
Taxable income (from PAT)	$PAT / (1 - t)$; EBIT = Taxable income + Interest
Financial leverage	PBIT / PBT
Distressed equity	$DCF \times (1-P) + \text{Distress value} \times P$
APV	Unlevered value + PV tax shield – Expected bankruptcy cost
Expected CF (distress)	$CF \times (1 - P(\text{distress}))$

16. Solved practical questions — pattern recap

Q	Key method & answer
1 ABC-XYZ (1.5 cr shares @ ₹400; CFs 250/300/400 @12%)	Market value = ₹600 cr; DCF = $250 \times 0.893 + 300 \times 0.797 + 400 \times 0.712 = \text{₹747.15 cr}$ (₹498.10/share) → range ₹600 cr - ₹747.15 cr (₹400-₹498.10/share)
2 Eagle Ltd.	PBT = $77/0.7 = 110$; $-8 + 10 = 112$; + new product profit ($70 - 20 - 12 - 10 = 28$) = 140; tax 30% = 42 → 98 L ; $\div 0.14 = \text{₹700 L}$. Earnings to equity = $98 - 13$ (pref. div) = 85 L $\div 50$ L shares = ₹1.70 $\times$ PE 10 = ₹17/share
3 ABC new sales strategy (20% growth $\times 3$ yrs, 15%)	FCFs negative in Y1-3 ($-720, -864, -1036.80$), Y4 = 2,419.20; PV($-1,961.79$) + PV of residual ($2419.2/0.15 = 16,128 \rightarrow 10,603.55$) = 8,641.76 vs. pre-strategy $1,400/0.15 = 9,333.33 \Rightarrow \text{₹691.57} \rightarrow \text{strategy not viable}$
4 H Ltd. & B Ltd. exchange ratio	NAV/share: H ₹285.71 (after deducting ₹300 cr contingent liability), B ₹48.46. Earnings cap/share: H ₹1,071.43, B ₹192.31. Fair value (weights 1:3): H ₹875 , B ₹156.35 → exchange ratio 0.1787 (H issues 0.1787 share per B share)
5 AB Ltd. & XY Ltd.	Share price ₹500 (avg of 570 & 430). DCF value 592.40 L, net assets 250 L → avg ₹421.20 L ; shares = $4,21,20,000/500 = \text{84,240}$; allocate on fully-paid equivalent (25 L): fully paid 67,392, partly paid 16,848
6 Hansel Ltd.	Gordon: $1800 = 54/(K_c - 0.09) \rightarrow K_c = 12\% \rightarrow$ book weight of debt 80%; correct weights 60 : 72 (E : D) → $K_c = \text{14.5455\%} \rightarrow$ value = $54/(0.145455 - 0.09) = \text{₹974.73 L}$
7 XYZ Ltd. (2-stage)	High growth: $K_e 16.9\%$, $K_d 9.1\%$, WACC 13%; stable: $K_e 14\%$, $K_d 9\%$, WACC 12%. FCFF Y1-4 = $56 / 67.20 / 80.64 / 96.77$; PV (13%) = 217.38; TV = $375.32 / (0.12 - 0.10) = 18,766 \rightarrow$ PV 11,503.56; Firm value ₹11,720.94 cr
8 WXY Ltd. (2-stage, WACC 15%)	FCF Y1-3 = $1342.50 / 1619.62 / 1953.47 \rightarrow$ PV 3,676.44; TV = $2680.13/(0.15 - 0.08) = 38,287.57 \rightarrow$ PV 25,174.08; Firm value ₹28,850.52 cr
9 Jatayu Ltd. EVA	Fin. leverage 1.5 = $\text{PBIT}/(\text{PBIT} - 40) \rightarrow$ PBIT 120; NOPAT 84; WACC = $14\% \times 300/700 + 7\% \times 400/700 = 10\%$; EVA = 84 - 0.10 $\times$ 700 = ₹14 L
10 RST Ltd.	$K_e = 8.5 + 1.36 \times 9 = 20.74\%$; K_d post-tax 7.70%; WACC 17.58% ; EBIT 40.11 L; EVA = -₹92,700
11 Tender Ltd.	PBT 21.43 L; EBIT 31.43 L ; EVA = $22,00,000 - 12,35,000 = \text{₹9,65,000}$; EVA dividend = ₹1.6083/share
12 Orange/Grape/Apple	WACC 13.52% / 15.225% / 17.95%; EVA 2,730 / 1,025 / -1,700 → Orange best (highest EVA, lowest WACC); EPS 1.30 / 1.45 / 1.43; price @PE 11 = 14.30 / 15.95 / 15.73; market cap 87,230 / 1,32,385 / 1,57,300 (PE should be risk-adjusted for differing leverage)
13 Delta Ltd.	Taxable income 25 L; EBIT 40 L ; EVA = $24 \text{ L} - 12.6 \text{ L} = \text{₹11.40 L}$; EVA dividend ₹4.56/share ; if no dividend, value <i>increases</i> (higher growth/EBIT)
14 XYZ Inc.	NOPAT 117; capital employed $605 \times 12\% = 72.60 \rightarrow$ EVA \$44.40 m ; MVA = market value 500 – equity fund 425 = \$75 m
15 Herbal Gyan	Include off-book patent (₹40 L): investment $140 \text{ L} \times 15\% = 21 \rightarrow$ EVA = 12 - 21 = -₹9 L
16 Constant Engineering	Capital = $200 + 400 + 100$ (patent) = 700; $K_e = 12\%$; WACC = $12(300/700) + 10(400/700) = \text{10.85\%}$; EVA = 84 - 0.1085 $\times$ 700 = ₹8.05 cr

Common traps

- Convert PAT → PBT using **(1 – t)** before adding interest to get EBIT.
- **Include unrecorded intangibles (patents)** in invested capital for EVA (Q15, Q16).
- For re-gearing beta use **market** weights, not book weights (Q6).

- In 2-stage DCF use **different WACC** for stable phase and discount TV with the **last high-growth-year** factor.
 - Adjust EBITDA/EBIT for **extraordinary / one-time items** before applying multiples (Illustration 1).
 - Present a **range** (min-max) and **comment on each method** in evaluation questions.
 - Theory Qs from the chapter: (1) *EVA vs MVA* (§6.3–6.4), (2) *Relative valuation* (§5).
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17. Quick theory Q&A (likely exam questions)

Q. Differentiate EVA and MVA.

EVA = NOPAT – WACC × invested capital: an *internal*, accounting-based measure of management efficiency over a period (return above hurdle rate). MVA = market value – invested capital: *external*, shows the market's view of cumulative value created; says nothing about share valuation. MVA is high for market darlings; EVA is the derived measure for discerning investors; in the short run they may be negatively correlated but converge.

Q. Explain relative valuation.

Comparative approach: derive a multiple (EV/EBITDA, EV/Sales, P/E, PEG...) from comparable firms and apply to target. Four steps – choose driver, choose ratio, choose comparables, iterate. Assumes efficient market, linear fundamentals-multiple link, comparable structure/risk/growth. Used to cross-check intrinsic value.

Q. Why can't traditional methods value start-ups? Table in §10.1.

Q. Going concern vs liquidation value? §8.

Q. How to treat ESG in valuation? §13 – through discount rate or (preferably) explicit cash-flow adjustments for E, S and G.

Q. Why is DCF unsuitable for distressed firms and what are the alternatives? §9.